

HUMAN PERFORMANCE GROUP CHAIRMAN'S
FACTUAL REPORT
(19 pages)



**NATIONAL TRANSPORTATION SAFETY BOARD
OFFICE OF HIGHWAY SAFETY
WASHINGTON, D.C. 20594**

HUMAN PERFORMANCE GROUP CHAIRMAN'S FACTUAL REPORT

A. ACCIDENT

Accident: Tunnel Ceiling Collapse
Location: Eastbound Interstate 90 connector tunnel, about 1000 feet west of the Ted Williams Tunnel, Boston, Suffolk County, Massachusetts
Vehicle #1: 1991 Buick 4-door passenger car
Operator #1: Passenger Vehicle
Date: July 10, 2006
Time: 11:01 p.m. local time
NTSB #: HWY-06-FH-024

B. HUMAN PERFORMANCE GROUP

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Senior Human Performance Investigator
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[REDACTED]

C. ACCIDENT SUMMARY

About 11:01 p.m. local time on Monday, July 10, 2006, a 1991 Buick passenger car was traveling eastbound in the left lane of the Interstate 90 connector tunnel to the Ted Williams Tunnel in Boston, Massachusetts. The vehicle was occupied by a 46-year-old male driver and a 38-year-old female passenger. A section of the tunnel's ceiling structure, consisting of ten 8-foot by 12-foot concrete panels and associated hardware, collapsed and struck the Buick. The Ceiling was held in place by multiple stainless steel threaded-rod inserted into holes in the roof of the tunnel and secured with an epoxy adhesive. A post-crash inspection revealed that 20 of the anchors had pulled out from the

¹ Not present during on-scene portion of the investigation.

roof of the tunnel. These anchors were used to attach eight metal brackets that supported a portion of the ceiling structure. Initially, two of the concrete panels fell onto the passenger vehicle crushing the right side of the roof. Following this impact, the passenger car came to rest against the north wall of the tunnel, and additional panels came to rest on the vehicle. As a result, the female passenger occupying the front right seat was fatally injured and the male driver received minor injuries.

D. DETAILS OF THE INVESTIGATION

The Human Performance factual investigation focused on the use of the adhesive anchoring system; specifically, contractual requirements, manufacturer's directions, contracting company procedures, and actual installation procedures were examined.

E. REQUIREMENTS IN THE CONTRACT

Modern Continental Construction Company, Inc. (MCC) was awarded the contract² for the I-90 Seaport Tunnel Access Finishes, which included the ceiling for the I-90 connector tunnel (the accident location). MCC was a contractor to the Massachusetts Highway Department. The installation of the adhesive anchoring system was described in Section 2 of MCC's contract.³ The section – "Installation Procedure & Quality Control Program Adhesive Anchoring System" – states that MCC intends to perform the installation in accordance with the manufacturer's guidance and will make provisions to provide the proper temperature of the adhesive and substrate. The installation procedure to be used is also specified, and is summarized below:

1. Locations of structural elements will be determined by survey and indicated on the substrate.
2. A template or the actual structural element will be used to locate each anchor on the substrate.
3. Holes will be drilled in the substrate, using a template to ensure proper orientation of the drill and drill bit. A drill stop will be used to ensure proper depth. Hole diameter will be determined in accordance with manufacturer's recommendations. MCC will confirm employees are equipped with the proper drill bits, anchors, and accessories for installation.
4. The drill hole shall be brushed clean, free of drilling debris and dust. The anchor will be inserted into the hole to ensure proper sizing.
5. On floor applications, the hole will be filled approximately half full with adhesive, starting from the rear of the hole. On wall and ceiling applications, a screen tube shall be used. After confirming the fit of the anchor in the hole, a screen tube shall be filled with epoxy, starting from the rear of the screen tube. The screen tube will be inserted into the hole

² Contract No. 99119-C09B2.

³ See Human Performance Attachment A: Section 2 of Contract General Provisions.

and seated at the required depth. An anchor seal plug will be inserted into the hole.

6. The anchor shall be free of dirt, oil, and foreign matter and inserted into the hole (through the plug on ceiling and wall applications) while being turned to ensure positive displacement of the epoxy. The anchor shall be inserted to proper embedment.
7. The anchor shall not be disturbed or loaded until after the manufacturer's minimum cure time.

In April of 1998, Addendum No. 12 was issued; part 3.07, "Installation of Adhesive Type Ceiling Anchorage System"⁴ contained additional directives regarding the installation of the adhesive anchors. Compliance with the manufacturer's instructions was again stressed and the testing procedure was specified. New instructions to follow when encountering reinforcing bars were provided and are summarized below:

D. Core Drilling Through Embedded Reinforcing Bars

1. The Contractor shall be allowed to core drill into the tunnel roof slabs and through the embedded reinforcing bars in accordance with the following procedures:
 - a. Should rebar interference be encountered during the course of normal initial drilling operations, the hole in question shall be identified with paint. The Engineer shall be notified and shall verify if the interference is a rebar or another object such as but not limited to a conduit.
 - b. Abandon the first hole and drill a new hole 3 inches away from the initial location. If there is an interference on the second attempt notify the Engineer who again will verify the type of interference.
 - c. If the first hole contained a rebar the Contractor may proceed with core drilling through the initial drilled hole upon receiving authorization from the Engineer.
 - d. Should the interference not be reinforcing bars the Engineer shall provide the Contractor with direction on how to proceed with the hole drilling.
 - e. Core drilling shall be perpendicular to the roof slab surface.
 - f. Maintain hole diameter as required by anchor manufacturer.
 - g. Core drill through entire section of embedded rebar. Provide additional adhesive and threaded rod when depth of hole exceeds typical depth.
 - h. Document all core drilling operations in a Core Drilling Report which shall include at a minimum the locations of all core drilled holes, the depth and diameter of all core drilled holes, and the number, size, and location of all embedded rebar which were cored. Include a plan which documents the location of each cored hole.

⁴ See Human Performance Attachment B: Design Specifications Section 723.480, 3.07.

- i. Fasteners required to anchor drilling equipment to tunnel roof shall not cut reinforcing steel or other embedded items.

F. MODERN CONTINENTAL CONSTRUCTION PROCEDURES

In a meeting with NTSB investigators, representatives and counsel for Modern Continental Construction described the procedures used when drilling the anchor holes and applying the adhesive (epoxy).⁵ According to MCC, Powers (Rawl) was selected as the supplier for adhesive from a list of three pre-approved suppliers in the contract.⁶ Product information for the adhesive and the installation procedure were submitted to Bechtel/Parsons Brinckerhoff (B/PB) and Gannett Flemming for review and approval. The drilling and installation procedure was described as follows:

- **Drill a hole using a core drill.** A water-cooled core drill equipped with a 5/8" diamond-tipped bit was used to drill a hole to a depth of 5" to 5 1/2". The drill's mechanical stop was used to ensure correct depth. The use of the core drill was specifically allowed by the contract and a diamond-tipped bit was specifically required by the contract.
- **Blow out the holes with compressed air.** An oil-free electric compressor was used to blow out the holes.
- **Clean the hole with a wire brush.**
- **Blow the hole out again.** Following this step, hole depth was measured and recorded in a spreadsheet.
- **Let the hole stand for 2 to 3 days.**
- **Return to the hole and insert a "red cap".**
- **Dispense and discard a small amount of adhesive.** This would ensure the adhesive was properly mixed.
- **Insert the nozzle through the red cap.**
- **Fill the hole at least halfway with epoxy.**
- **Insert a pre-cut rod into the hole.** The rod was inserted through the red cap and was twisted as it was pushed into the hole. The rod was inserted into the hole until contact with the bottom (bottomed out).
- **Hold rod in place until adhesive sets up.** The length of rod remaining outside the hole was measured.
- **Allow adhesive to cure for 2-3 days.**
- **Rod/adhesive installation is tested.** Per the contract, 100% proof testing was performed.

According to MCC, a representative from Powers was on-site on July 29, 1999 during the installation of 50 anchors in the High-Occupancy Vehicle (HOV) lanes to provide direction, explanation, and assistance as needed. A representative from Powers

⁵ See Human Performance Attachment C: Meeting Notes, September 27, 2006.

⁶ An examination of the contract indicates there was not a pre-approved list of suppliers in the Ceiling Finishes section of the contract, but there was such a list in the Miscellaneous Steel Specification section.

also came to the tunnel on October 21, 1999, to examine the anchor failures in the HOV area.

G. EPOXY MANUFACTURER'S INSTRUCTIONS

Information gathered by the Attorney General's Office indicated Modern Continental Construction purchased an adhesive anchor system (epoxy and bolts) from Newman Renner Colony, including epoxy sold under the name NRC 1000 Gold Premium Epoxy. This epoxy was formulated by Sika Corporation and sold to Powers Fasteners, which, in turn, packaged it for sale as Power-Fast Epoxy Injection Gel and as NRC 1000 Gold Premium Epoxy. In submitting materials on the product for review, MCC submitted documentation on Power-Fast Epoxy Injection Gel.

According to literature on the product,⁷ Power-Fast Epoxy Injection Gel is a two-component structural epoxy designed for use in anchoring threaded rods, bolts, reinforcing bars, and smooth dowels into concrete and other solid base materials. It can also be used to anchor into hollow masonry materials, to bond steel and cured concrete to cured concrete, for pick proofing applications, for surface sealing cracks, and for placing injection ports. The product has a mixing ratio of 1:1 by volume. Power-Fast Epoxy Injection Gel was available in either a fast or slow setting version to provide the installer with a choice of gel and curing times.⁸

Topics addressed under Adhesive Anchor Installation Guidelines in the manual include anchor holes and use of adhesives in cold weather. With respect to anchor holes, Powers Fasteners states that their adhesives are designed to be installed in holes drilled with carbide bits meeting American National Standards Institute (ANSI) Standard B212.15 unless otherwise specified. The guidelines go on to state that if alternate bit types, such as diamond tipped core bits, are used the tip tolerance should be within the ANSI range. The guidelines note that diamond tipped bits drill a hole with very smooth walls, causing some anchor types to slip and fail prematurely. Powers Fasteners also states that adhesive anchors should be tested in diamond-cored holes prior to use.⁹

When using the product in cold weather, Powers recommends the adhesive be conditioned to a minimum temperature of 60° F (degrees Fahrenheit) unless otherwise recommended for a specific product. Conditioning requires the adhesive be maintained at the required temperature for a sufficient time to ensure that the entire cross section of the container is brought to temperature. The general installation guidelines also

⁷ The Second Edition of the Powers-Rawl/Powers Fastener Design Manual has a copyright date of 1997. The copyright date of the Third Edition of the manual could not be determined, making it unclear which was in effect at the time the epoxy was selected and used in the tunnel. However, an October 29, 1999 letter from Powers Fasteners refers to the Second Edition, indicating it was likely still current at that time. Therefore, staff references the Second Edition in this report.

⁸ See Human Performance Attachment D: Pages from Powers-Rawl Fastening Systems Design Manual, Second Edition (p. 146-149).

⁹ See Human Performance Attachment D: Pages from Powers-Rawl Fastening Systems Design Manual, Second Edition (p. 145).

recommend not using epoxies in base materials having a temperature less than 40° F unless job site performance tests are conducted.¹⁰

Section 22.6.1 of the 1997 manual, “Allowable Load Capacities”, describes allowable loads and safety factors for anchors. The section states, in part:¹¹

“The standard established by the industry is to reduce the ultimate load capacity by a minimum safety factor of 4 to calculate the allowable working load. For example, an anchor which has an average ultimate tension load capacity of 12,000 pounds would have a maximum allowable working load anchor of 3,000 pounds. Critical applications such as overhead applications or dynamic loading may require safety factors of 10 or higher. The allowable loads are recommendations, however, local building codes should be consulted to determine the required safety factors.”

Specific installation guidelines for “Power-Fast” can be found in Powers’ literature.¹² For solid base materials, the instructions are:

- “Drill a hole to the size and embedment required. The tolerances of the drill bit used should meet the requirements of ANSI standard B212.15.
- Blow the hole clean with compressed air, brush the hole, and blow it clean again. Holes should be clean and sound. They may be dry or damp, but should be free of standing water or frost. If using reinforcing bars, be sure the bar will fit into the drilled hole. If a larger hole is required, the diameter should be as close as possible to the diameter of the reinforcing bar.
- Be sure to properly balance the mixing nozzle prior to dispensing and when changing cartridges or sausages. Fill the hole approximately half way with epoxy starting from the rear of the hole. Slowly withdraw the static mixing nozzle as the hole fills to avoid creating air pockets within the hole.
- Push the threaded rod or reinforcing bar into the hole while turning slightly to insure positive distribution of the epoxy. Be sure the rod is fully seated at the bottom of the hole and that some epoxy has flowed from the top of the hole. The threaded rod or reinforcing bar used should be free of dirt, grease, oil, or other foreign material.

¹⁰ See Human Performance Attachment D: Pages from Powers-Rawl Fastening Systems Design Manual, Second Edition (p. 145).

¹¹ See Human Performance Attachment D: Pages from Powers-Rawl Fastening Systems Design Manual, Second Edition (p. 67).

¹² See Human Performance Attachment D: Pages from Powers-Rawl Fastening Systems Design Manual, Second Edition (p. 159).

- Allow the epoxy to cure for the specified time prior to applying any load. Do not disturb or load the anchor until it is fully cured.”

Section 52.10.2 of the Second Edition Manual covers the installation of Power-Fast anchors in an overhead position.¹³ The section states that Power-Fast epoxy injection gel can be used for the installation of threaded anchor rod overhead. The overhead anchor holes should be prepared using the standard procedure with the following exceptions:

- After the anchor hole is drilled and cleaned, insert the proper size anchor seal plug into the hole;
- Insert the Power-Fast mixing nozzle through the seal plug and fill the hole half way starting from the bottom or rear of the hole;
- Insert the threaded rod through the seal plug, turning it slightly to ensure proper distribution of the epoxy; and
- Be sure the anchor rod is fully seated at the bottom of the hole and some epoxy has flowed from the opening in the seal plug.

H. ACTUAL INSTALLATION PROCEDURES

In an interview with NTSB investigators,¹⁴ one of the laborers responsible for drilling the holes and installing the adhesive anchors described the procedure as follows:

- A two-man crew would use a scissors-lift to raise themselves and the water-cooled drill to the ceiling of the tunnel.
- They would line the drill up with a mark placed on the ceiling by the surveyors, secure the drill to the ceiling, and drill the hole using a core drill with a diamond-tipped bit.
- Depth of the hole was obtained through markings on the drill bit and/or a drill stop.
- Once the hole was drilled, they lowered the drill and checked the depth of the hole with a tape measure.
- The crew would drill holes for several days or a week before cleaning the holes.
- The holes were cleaned using an “iron” (round wire) brush. The brush was inserted in the hole and moved along its length several times.
- Next, compressed air was used to blow out the holes. The hole would be allowed to dry two or three days.
- Two men were used when installing the adhesive anchors; one would inject the epoxy into the hole and the other would insert the anchors.

¹³ See Human Performance Attachment D: Pages from Powers-Rawl Fastening Systems Design Manual, Second Edition (p. 164).

¹⁴ See Human Performance Attachment E: Narratives of Investigative Interviews.

- When injecting the epoxy, a “red cap” was inserted and the injection gun was inserted into the hole through the “red cap”.
- The epoxy nozzle was “bottomed out” in the hole and was slowly withdrawn while dispensing epoxy. The hole was filled with epoxy.
- After the hole was filled with epoxy, a pre-cut anchor (bolt) was inserted into the hole through the red cap and was twisted as it was inserted. Any excess epoxy was wiped away and the portion of the bolt extending from the hole was checked for length.

When asked, the laborer indicated that he would run some of the epoxy out of the gun when using a new tube to ensure he was using properly mixed epoxy. The laborer believed the epoxy being used was the fast-set version, but also stated he was not sure. When asked about bolts failing during testing, the laborer stated he did have to re-drill holes and re-insert epoxy. The laborer indicated the extent of the training he received on drilling and using the epoxy consisted of instruction, demonstration, and observation on the first day or so.

A second laborer interviewed by NTSB investigators¹⁵ provided a similar description of the drilling process. The second laborer was able to recall that the air compressor used was an electric compressor. The second laborer could only recall one bolt failing during testing, resulting in the surveyors moving the entire plate. The second laborer stated he was trained to use the drill and the epoxy by the first laborer.

NTSB investigators also interviewed the labor foreman.¹⁶ His account of the drilling and adhesive process was similar to the accounts provided by the two laborers. The foreman was able to add that a limited amount of epoxy was maintained on-site; when that ran out, he was responsible for getting more from the Boston company office. Staff from the Boston office ordered additional cases of epoxy as needed. When asked about the procedure when a bolt failed the pull-out test, the foreman indicated the bolts were removed, the holes were drilled out (either with the core drill or a hammer drill), and fresh epoxy and a new bolt were inserted in the hole. When asked about the temperature in the tunnel, the foreman stated the temperature in the tunnel never dropped below 45 degrees.

I. SELECTED ENTRIES FROM FIELD ENGINEER’S DAILY REPORT

An examination of the Field Engineer’s Daily Report¹⁷ revealed a number of entries related to the installation of the adhesive anchor system. These entries are summarized below:

¹⁵ See Human Performance Attachment E: Narratives of Investigative Interviews.

¹⁶ See Human Performance Attachment F: Transcript of Investigative Interview.

¹⁷ See Human Performance Attachment G: Excerpts from Field Engineer’s Daily Report

- **June 10, 1999:** Prototype drilling of holes for ceiling support anchor bolts in the portal area, eastbound I-90. Core drill supplier (Hilti) present. Eight holes drilled, no rebar encountered.
- **June 22, 1999:** Laborer and labor foreman continuing to core drill ceiling anchor holes. Twenty-six holes drilled, with five encountering rebar and shifted 3" per specs.
- **June 23, 1999:** MCC crew continued drilling holes in ceiling of I-90 EB for ceiling support system. Over course of day, 20 holes successfully drilled and eight abandoned due to rebar hits.
- **June 24, 1999:** MCC continued drilling in I-90 EB. By end of day, 6 holes successfully drilled and 11 abandoned due to rebar hits.
- **June 25, 1999:** A total of 23 new holes drilled; also, crew returned to four holes abandoned due to steel encounter. On advice of Hilti representative, drill rig adjusted to allow drilling through rebar in approximately 10 minutes. A total of 117 holes have been drilled with steel encountered at 34 locations (29%).
- **June 28, 1999:** Total of 23 holes drilled; 18 new and 5 re-drills where rebar had been hit. To date, a total of 140 holes drilled with 35 rebar hits (25%).
- **June 29, 1999:** Twenty-seven holes attempted today; at three of the attempted holes, a rough roof surface prevented the formation of a vacuum.
- **June 30, 1999:** A total of 17 holes attempted today. At two of the holes, a rough roof surface prevented the formation of a vacuum. Of the remaining 15 holes, two were previous rebar hits. Two of the 13 new holes encountered steel. To date, 180 holes have been drilled with rebar at 40 locations, a hit-ratio of 22%.
- **July 1, 1999:** Eleven holes drilled in the Eastbound bore. A total of 13 holes drilled in the HOV roof, for a total of 24 holes. About 5 or 6 of these encountered steel; precise number/location of steel hits not obtained.
- **July 2, 1999:** Drilling continues in HOV area, with a total of 12 holes drilled (two reported steel hits). Overall, 4 hits in 26 holes or 15%. Steel hit rate far below the usual rate of 25-30% in Eastbound lanes and Ted Williams Tunnel. Re-bar core plug found on ground, although no drill-through reported. Magnetic test of all holes drilled to date may be needed to verify MCC's steel encounter report. MCC not marking steel hit locations as required by contract. Design layout of the anchor bolt pair at HOV line A, Row 33 falls in an electrical pullbox. Bolts relocated 2 feet west of design location.
- **July 6, 1999:** Two laborers drilling, with 14 attempts, 5 steel hits, and 9 successful. Drilling rig malfunctioned about 2:00, no further drilling today.
- **July 7, 1999:** New drill brought in to replace drill that failed yesterday; new drill failed at about 11:00. Third drill brought in and a new water collar affixed to prevent water from entering drill casing (apparent cause of drill failures). Third drill functioned successfully for rest of day; 11 holes drilled with no reported steel hits. There were 2 surveyors doing ceiling anchor bolt layout in the HOV bore, just ahead of the drillers.
- **July 8, 1999:** MCC continued drilling holes in the HOV lane ceiling; 30 holes drilled with no rebar hits mentioned.
- **July 9, 1999:** Thirty-two holes drilled with one steel hit. Two surveyors were doing lay-out in the WB bore.

- **July 12, 1999:** Drilling continues in HOV bore near East Portal; 37 holes drilled with no steel hits. Of last 69 holes drilled, only one steel hit reported. Two surveyors were doing ceiling anchor bolt layout in the westbound (WB) bore.
- **July 13, 1999:** Seventeen (17) holes drilled in HOV bore with 3 steel hits.
- **July 14, 1999:** Twenty holes drilled with no reported steel hits. Laborers were going to remain late to finish the last two holes in HOV.
- **July 15, 1999:** Nineteen (19) holes drilled in eastbound (EB) bore with one steel hit. Surveyors were laying out anchor bolt locations in the EB bore.
- **July 16, 1999:** In the EB bore, 18 holes were drilled with 2 steel hits.
- **July 19, 1999:** A total of 43 holes drilled in EB bore with no steel hits. Surveyors laying out anchor bolts holes in the WB bore.
- **July 20, 1999:** A total of 32 holes drilled in the EB and WB bores. EB drilling complete at 10:30. No steel hits reported. Two surveyors were doing layout in the HOV.
- **July 21, 1999:** Two MCC laborers continued drilling holes; 64 holes drilled with no rebar hits.
- **July 22, 1999:** Two MCC laborers continued drilling holes; 40 drilled with no rebar hits.
- **July 23, 1999:** Conducted magnetic survey of bolt holes in HOV; 50 random locations reported as not encountering steel selected. Magnetic inspection revealed 32 of 50 did in fact contain rebar. Will compare results with log of drilling results. MCC laborers continued with drilling; 30 holes with 5 rebar hits.
- **July 28, 1999:** One laborer was setting anchor bolts in Mock-up area of HOV; bolts installed without screens to hold the epoxy. A total of 30 bolts set.
- **July 29, 1999:** On laborer setting bolts in HOV; bolts installed without screens and without blowing the holes out with compressed air as per the manufacturer's recommendations. A total of 52 bolts were set. Safety officer has required installers to wear Tyvek clothing per manufacturer's recommendation.
- **July 30, 1999:** Anchor pull out testing conducted in HOV mock-up area. A total of 36 bolts tested with 3 failures. Two additional bolts were not epoxied in place. MCC laborers drilled and set anchor bolts in HOV at locations that had been skipped. After placing bolts in the skipped areas, they moved to line 12 (MCC layout) and continued drilling. A total of 17 holes drilled today.
- **August 2, 1999:** Representatives from CONAM Inspection Inc. began testing. A total of 35 bolts were tested with three failures.
- **August 3, 1999:** One laborer setting anchor bolts in the EB bore. A total of 46 bolts installed. No testing of anchor bolts performed today.
- **August 4, 1999:** Technicians from CONAM were testing anchor bolts in HOV and EB, including locations that had previously failed and new bolts were set. A total of 34 bolts in EB (2 failures) and 11 in HOV (1 failure). One MCC laborer was blowing out bolt holes with compressed air and setting anchor bolts in the EB bore.
- **August 5, 1999:** CONAM technicians tested 14 bolts in HOV, including locations that had previously failed; three (3) failures occurred. Failed bolts were removed and will be replaced and re-tested. One MCC laborer was blowing out bolt holes with compressed air and setting anchor bolts in the EB bore.

- **August 9, 1999:** A total of 25 plates and 17 hanger assemblies were installed in the HOV area. Two technicians from CONAM performed anchor bolt pull out tests. In the HOV, four (4) bolts were tested and 2 failed; one bolt (Line 'B', Row 30, North) passed the pull out test but was observed to turn when a nut was tightened on it. The other (Line 'C', Row 32) held the load but was displaced ½ inch. In EB, 40 bolts were tested and there were 3 failures (Line 'A', Row 25, North; Line 'B', Row 24, South; and Line 'A', Row 21, South). The failed bolts were marked and will be replaced and re-tested.
- **August 11, 1999:** Two laborers and one labor foreman drilled holes for ceiling support in the mock-up area (4 holes).
- **August 12, 1999:** One ironworker installed four (4) anchor bolts. Two technicians from CONAM performed pull out tests; anchor bolt in HOV column line C35 failed; will be replaced and re-tested.
- **August 13, 1999:** Technicians from CONAM performed pull out testing in HOV and EB bore; a total of 40 bolts tests with 1 failure (HOV column line A21). Failure marked and will be replaced, re-tested.
- **August 16, 1999:** Two technicians from CONAM performed pull out tests in HOV area column lines C37, C38, North Side, and South Side and retested C32 and C35. Six total tests performed; all passed.
- **August 18, 1999:** One laborer installing ceiling anchor bolts, I-90 EB.
- **August 23, 1999:** One MCC laborer blowing ceiling holes WB with compressed air.
- **August 30, 1999:** Ironworkers from MCC drilling and setting anchor bolts into pre-cast ceiling panel and roof slab (HOV078, HOV079). Bolts were installed without blowing the holes out with compressed air per Manufacturer's recommendations. Bill and Scott from MCC notified.
- **August 31, 1999:** Ironworkers from MCC drilling and setting anchor bolts into pre-cast ceiling panel and roof slab (HOV078, HOV079). Bolts were installed without blowing the holes out with compressed air per Manufacturer's recommendations. Bill and Scott from MCC notified. Future installations are to be done correctly.
- **September 7, 1999:** Two technicians from CONAM performed pull out tests in the EB 4A2 area; 58 bolts tested with one failure. The failed bolt was removed and will need to be replaced & re-tested.
- **September 8, 1999:** Two technicians from CONAM performed pull out tests in the EB 4A2 area; 48 bolts tested with 2 failures. The failed bolts were removed and new bolts will need to be installed and tested.
- **September 17, 1999:** Two technicians from CONAM performed pull out tests in the WB 4A2 area; 58 bolts tested with 2 failures. Failed bolts removed and will need to be reinstalled and retested.
- **September 20, 1999:** One MCC laborer drilling and installing adhesive anchor bolts in the EB 4A2 area. A total of six (6) bolts were drilled and set and will need to be pull tested prior to hanger installation.
- **September 22, 1999:** Two technicians from CONAM performed pull tests on 56 bolts in the WB 4A2 area; there were no failures.
- **September 24, 1999:** Two technicians performed pull out tests in the WB 4A2 area. By COB they had tested 56 bolts with no failures.

- **September 28, 1999:** Two laborers were drilling and installing adhesive support anchor bolts in the HOV 4A2 area that had been previously skipped (Line 'C', Rows 12 & 13).
- **October 13, 1999:** Met with representatives from MCC and Powers Fasteners, Inc. to discuss problem of anchor bolts "creeping" out over time. Showed Powers representative the problem and reviews installation and testing procedures, along with module erection process.
- **October 21, 1999:** Field meeting held with representatives from Powers, MCC, and B/PB. Purpose of meeting was to discuss and determine the cause of the pull out of five of the ceiling anchors in the mock-up at module HOV078. Installation and testing procedures for the ceiling anchors were reviewed. Several areas of potential concern were discussed: 1) Installation temperature - It was determined that this was not an [...] ¹⁸.
- **November 5, 1999:** Two technicians from CONAM performed ceiling anchor bolt pull out tests in the EB and WB bores of the 4A2 area. Forty (40) bolts were tested with one (1) failure.
- **November 6, 1999:** Two CONAM technicians performed pull out testing in the WB area. A total of 36 bolts were tested with no failures.
- **November 9, 1999:** Selected 66 bolts (one-third of installed bolts) for torque testing. Twenty-two (22) were taken from each of the three longitudinal lines of installed bolts. All tested bolts required approximately 1/8 turn to "click" a torque wrench set to 75 ft-lbs. Manufacturer's recommended torque for 5/8" bolts is 50 to 90 ft-lbs; hence 75 ft-lbs is acceptable.
- **November 12, 1999:** Two technicians from CONAM conducted pull tests on two bolts in WB which were previously skipped, and both passed. Technicians then went to mock-up area to pull out a bolt which had passed the pull-out test (passed on 7/30/99) but had been slipping out over the last several months (a survey on 10/08/99 found five bolts were slipping, with this bolt showing the most displacement at 9/16"). When standard pull-out load (3.25 kips) was applied today, the bolt held for a few seconds, then began to pull out with almost no resistance. Upon examination, the hole appeared to lack sufficient epoxy to fill the hole. A significant amount of dust was adhered to the epoxy, a sign the hole had not been completely cleaned. The epoxy near the embedded tip of the bolt was brittle and crumbled easily, a sign of improper mixing. The epoxy near the exposed end appeared to be fully cured.
- **November 15, 1999:** Verified torque of bolts in EB 4A2 area; 64 bolts (25% of installed) or 16 from each of the longitudinal lines randomly selected. All tested bolts found to require 1/8 and 1/4 turn to "click" a torque wrench set to 75 ft-lbs. This value falls within manufacturer's recommended range of 50 to 90 ft-lbs.
- **November 17, 1999:** Labor foreman and two laborers staying late to drill and set six anchor bolts (three sets of two) in the ceiling at EB, Line A, adjacent to UR25. Six ironworkers, one operator, and one teamster hung modules EBD082A, EBD081A, EBD081B, EBD080A, ¹⁹ and EBD080B.

¹⁸ The page ends here; no further text is available in the copy provided to staff.

¹⁹ Modules EBD081A and EBD081B were the accident modules.

- **November 18, 1999:** Two laborers were drilling and installing adhesive anchor bolts in EB bore at approximately Sta 68+00. Labor foreman stated crew would be staying late to complete this task so the installed bolts could be tested on Monday (11/22) and not delay the ceiling hanging activity.
- **November 22, 1999:** Technicians from CONAM performed bolt pull-out tests on bolts installed last week in EB, approximately Sta 67+00. By COB they had tested 22 bolts with one failure. The failed bolt was removed and will need to be replaced, retested.
- **November 29, 1999:** Two laborers were drilling holes for adhesive anchor bolts at EB module 81²⁰ where shifting the originally installed bolts to miss steel adversely affected module alignment.
- **December 1, 1999:** Two laborers were drilling holes for ceiling anchor bolts in EB bore at approximately Sta 70+00.
- **March 13, 2000:** Technicians from CONAM tested bolts in the 4A2 area of HOV, east of mock-up. By COB they had tested 72 bolts with two failures. The failed bolts held the test load of 6350 lbs when it was first applied. The testing procedure requires the load to be held for two minutes; during the period, both bolts “popped” loosed. Both came out almost perfectly clean, with almost no cured epoxy clinging to either one. New bolts will be installed at the two locations and retested. This completes testing in the HOV 4A2 area, except for the bolts which are currently supporting the mock-up ceiling panels, which still need to be tested to the higher (6350 lb) load.

J. DISPLACEMENT OF ANCHORS IN THE HOV AREA

In early October 1999, it was noted that anchors in the HOV “mock-up” area²¹ had displaced, creating gaps in the ceiling hanger plate to CIP ceiling interface. On October 6, 1999, the issue was added to the agenda for a technical issues meeting scheduled for October 8, 1999.²² A memorandum from MCC dated October 7, 1999, indicated they noted no movement in August, noted small movement (1/16”) of an inch on September 9, 1999, and noted worst-case movement of ½” in early October 1999.²³

Minutes from the October 8 meeting state that the importance of properly adjusting all hangers so that each is taking its portion of the load prior to lowering the lifting device was discussed, with MCC stating that this had been done. MCC indicated that the vendor of the epoxy (Powers Fasteners) had been notified and would be visiting the job site to examine the problem. The minutes state that it was requested and expected that MCC would do what was necessary to determine the cause of the problem, how to remedy it, and how to prevent it from recurring. The calculations were to be re-visited to

²⁰ The accident module.

²¹ Ceiling module HOV079.

²² See Human Performance Attachment H: HOV Anchor Displacement Correspondence (e-mail dated October 6).

²³ See Human Performance Attachment H: HOV Anchor Displacement Correspondence (Letter C09B2-114).

verify the ceiling anchor system was properly designed and not overloaded. This agenda item was classified as “OPEN”.²⁴

A letter dated October 12, 1999 from Bechtel/Parsons Brinckerhoff indicated several potential causes for the anchor displacement, including:²⁵

- Loose turnbuckles inducing excessive loading;
- Releasing the lifting system too quickly, causing excessive stress;
- Releasing the lifting system prior to uniform loading of all turnbuckles; and
- Failure to properly prepare & clean the drilled holes.

The letter goes on state that MCC is to:

- Contact the supplier of the adhesive anchor bolt system, review the installation of the failed anchors, and obtain a manufacturer’s recommendation;
- Confirm that all ceiling hangers are correctly loaded and have uniform load; and
- Investigate, with the manufacturer, the possibility of retesting anchors to design loading after the supports are properly adjusted.

B/PB will be responsible for confirming that the ceiling load calculations are correct.

The Field Engineer’s Daily Report for October 13, 1999 notes that the Field Architect met with representatives from MCC and Powers atop the mock-up ceiling to discuss the anchor displacement. The Powers representative was shown the displaced anchors and the anchor bolt installation and testing procedures were reviewed, along with the module erection process. The Powers representative did not offer any conclusions or analysis at that time, instead deferring until he could consult with Powers’ engineering staff.²⁶

An October 19, 1999 memorandum from MCC to Powers²⁷ provided Powers with the testing procedure, a sketch of the mock-up panel area, and the testing reports for the anchors in question. The memorandum also asked a number of general questions:

- Is any movement in the anchor after loading considered typical?

²⁴ See Human Performance Attachment H: HOV Anchor Displacement Correspondence (Meeting Minutes, Friday, October 8, 1999; File No. CN-9.09B2.28.10).

²⁵ See Human Performance Attachment H: HOV Anchor Displacement Correspondence (Letter C09B2-0350).

²⁶ See Section I of this report and Human Performance Attachment H: HOV Anchor Displacement Correspondence (Field Engineer’s Daily Report, October 13, 1999).

²⁷ See Human Performance Attachment H: HOV Anchor Displacement Correspondence (October 19 Memorandum).

- Does any movement indicate an anchor failure?
- If there was a problem during installation (i.e. dust, bad epoxy mix, etc.) could the anchor pass a proof test? Would an installation problem manifest itself immediately, or could it develop over time?
- At what point in loading would the anchors become overstressed? In overstressing an anchor, is it typical to see epoxy failure or anchor elongation first?
- Loading of the anchors in the final condition is required to be at a minimum 4,000 lb. If the ceiling system loads were non-uniformly distributed among the anchors, could the anchors have become overstressed? Does the ceiling system weigh enough to cause overstressing in the anchors? Ceiling module is 40' x 12', weighs approximately 14 tons (28,000 lb), and is supported by a hanger system, attached 5' on center.

In a letter to MCC dated October 29, 1999, Powers indicates their representative checked the torque of nuts on both displaced and non-displaced anchors and got torque readings of 120 foot-pounds, a value in excess of the published maximums (50-90 foot-pounds) for a 5/8" diameter Power-Fast adhesive anchor. The letter goes on to state:

- If an appropriate factor of safety is applied to the ultimate load capacity of the adhesive, then displacement should not occur provided that proper installation procedures were employed.
- Any movement in the anchor may be an indication that the allowable load capacity of that particular installation has potentially been exceeded. When displacement becomes more significant, the anchor approaches its ultimate failure load capacity.
- If a problem occurred during installation, it is possible an anchor could pass a proof load test provided a large enough safety factor was employed.
- Adhesive anchoring problems typically reveal themselves immediately in a testing environment and do not manifest themselves over time. If the problem were dust in the hole, the anchor probably would have moved during the proof loading procedure. If not properly mixed, the adhesive typically would not fully cure and remain somewhat tacky. This would also be apparent during the proof loading procedure.
- Overstress is a term more often associated with the applied torque to an adhesive anchor. The condition where the nuts have possibly been overtightened is described as an overstressed condition. In an ultimate adhesive anchor failure condition, the mode of failure can either be base material failure or more commonly, bond failure at the interface of the adhesive and the walls of the anchor hole. Anchor rod elongation is more likely to occur in deeper embedment depths (greater than 5") where the strength of the anchor rod material has effectively been developed.
- Since Powers Fasteners, Inc. is not the designer of this ceiling hanger plate assembly, an answer to this final question cannot be provided. Powers is not aware of the different loads that may be acting on these

anchors. In addition to dead loads, live loads such as wind, equipment and, personnel that potentially affect the applied load capacity should be considered.

- While Powers believes the existing anchors subjected to torques exceeding their recommended range may be acceptable, the only way to verify their suitability is by performing a proof load test.

In a letter dated November 8, 1999, MCC disagrees with B/PB that improper ceiling module installation and anchor hole preparation are potential causes for the observed anchor movement. MCC states that if improper anchor installation (i.e. drill hole preparation, epoxy installation, etc.) were the problem, it is likely that the anchors would have failed the proof test. As they all passed, it is not likely that installation procedures were the cause of the anchor movement. MCC goes on to state that the ceiling modules were erected in accordance with the approved procedure in the presence of the Engineer; the assertion that the lifting devices were removed too quickly is unfounded.²⁸

The correspondence continues with a letter to MCC dated November 11, 1999.²⁹ In this letter, B/PB disagrees with MCC's position that all elements of the ceiling have been installed in accordance with the contract, citing over-torqued anchors and non-uniform loading. The letter requests that MCC confirm no over-torquing has occurred on previously installed plates before commencing new ceiling installations.

According to the Field Engineer's Daily Report, torque readings were taken on a sampling of anchor bolts on November 9, 1999 and November 15, 1999. All anchors checked on these days had torque readings within the acceptable range.³⁰

On December 2, 1999, Deficiency Report (DR) Number 001 – Adhesive Anchor Failures was sent to MCC for review and response.³¹ On the same day, MCC sent a response detailing their corrective actions to address the issue.³² As part of this response, MCC included a letter from Powers describing a removal and replacement procedure. On December 27, 1999, B/BP denied MCC permission to proceed with the replacement, requiring MCC to submit documentation on the epoxy and specifying a tension test at 125% of the Allowable Tension, or 7900 lbs.³³ On January 7, 2000, B/PB directed MCC to proceed with the replacement and re-test to the Allowable Tension value, or 6350

²⁸ See Human Performance Attachment H: HOV Anchor Displacement Correspondence (Letter C09B2-0139).

²⁹ See Human Performance Attachment H: HOV Anchor Displacement Correspondence (Letter C09B2-0409).

³⁰ See Section I of this report and Human Performance Attachment H: HOV Anchor Displacement Correspondence (Field Engineer's Daily Report, November 9 and November 15, 1999).

³¹ See Human Performance Attachment H: HOV Anchor Displacement Correspondence (Letter C09B2-0436).

³² See Human Performance Attachment H: HOV Anchor Displacement Correspondence (Letter C09B2-0155).

³³ See Human Performance Attachment H: HOV Anchor Displacement Correspondence (Adhesive Anchor Repair Procedure, 12/27/99).

lbs.³⁴ After some discussion of the responsibility for re-testing previously approved anchors, MCC was given authorization to proceed with the repairs. The final procedure specified a revised proof-testing load of 6350 lbs tension. After completing the revised test procedure in the HOV mock-up area, MCC was to provide the test results to the project Engineer and await further direction.³⁵

An August 31, 2000 memorandum summarizes the re-testing of the anchors at the higher load of 6350 lbs. It states that 187 anchors were tested and 19 failed at the higher load. Each of the failures was replaced and re-tested; one of these failed and was again replaced and re-tested and passed. Only two failures were observed outside the “mock-up” area; the memorandum states this could be due to better installation techniques used as a result of going through a “learning curve” in the “mock-up” area. The memorandum suggests the better techniques include: cleaning the holes out better, not using the first epoxy out of the gun, and not over-torquing the nuts when installing the ceiling plates. The final recommendation is that DR#1 be closed.³⁶

K. NTSB INSPECTION OF ANCHORS AND HOLES

As part of the investigation, staff from the NTSB Materials Laboratory completed a number of examinations of aspects of the adhesive anchor system, including measuring anchor displacement; taking core samples; and examining anchors and holes from both the accident location and other areas of the connector tunnel.³⁷

In an effort to understand the mechanics of the installation procedure and empirically determine the effects of deviating from the recommended installation procedure, NTSB investigators asked staff from the Federal Highway Administration (FHWA) to drill a number of holes in an overhead concrete slab using the same type of drill and drill bits used at the accident site. Similar anchors were inserted into the holes and held in place with Power Fast epoxy. After sufficient curing time, the anchors were subjected to pull testing. In addition to performing “ideal” installations following the manufacturer’s directions, FHWA staff also made a number of installation “errors”, such as failing to clean the holes, installing anchors in damp holes, reducing the length of the mixing nozzle, storage of the epoxy at elevated temperatures before installation, and using epoxy mixed at the wrong ratio of resin to hardener. FHWA staff also installed a series of anchors in a downward orientation for comparison, and studied the effects of bolt torque at three levels.

³⁴ See Human Performance Attachment H: HOV Anchor Displacement Correspondence (Letter C09B2-0499).

³⁵ See Human Performance Attachment H: HOV Anchor Displacement Correspondence (Adhesive Anchor Repair Procedure, 2/16/00).

³⁶ See Human Performance Attachment H: HOV Anchor Displacement Correspondence (Memorandum dated 8/31/00).

³⁷ Please see Materials Lab Reports.

END OF FACTUAL REPORT

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LIST OF ATTACHMENTS

Attachment A: Section 2 of Contract General Provisions
Attachment B: Design Specifications Section 723.480, 3.07
Attachment C: Meeting Notes, September 27, 2006.
Attachment D: Pages from Powers-Rawl Fastening Systems Design Manual, Second Ed
Attachment E: Narratives of Investigative Interviews
Attachment F: Transcript of Investigative Interview
Attachment G: Excerpts from Field Engineer's Daily Report
Attachment H: HOV Anchor Displacement Correspondence